Replication of the autobiographical-memory tests in Sorenson (2015)

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Abstract

Do adults with major depression give fewer specific autobiographical memories than adults who are not depressed, and do the printed tests in Sorenson (2015) follow from the printed means, standard deviations, and sample sizes? Sorenson reports a trend toward fewer specific memories in the depressed group, $t(35) = 1.96$, $p = .058$, no comparable impairment on a paired-associate recall test, and lower right prefrontal activity during successful image retrieval, with that cluster correlated with autobiographical specificity at $r = .46$. The tests fixed by the printed moments were recomputed in R, and the printed questionnaire means were checked as averages of integer scores. The specificity trend is recovered at the stated final sizes, 20 and 17: the moments give $t = 1.95$, the printed 1.96 lies in the half-digit band 1.93 to 1.98, and the two-tailed p-value of $t(35) = 1.96$ is .058. The Beck depression statistic 10.94 matches the recruited sizes 20 and 19, while the printed degrees of freedom 35 match the final sizes, where the same moments give $t = 10.36$. The negative-mood statistic 4.5 lies outside the half-digit band at 20 and 17 and is recovered at 19 and 18, the sizes at which those mood means are possible averages of integer scores. The right prefrontal cluster was not recomputed. No images or region summaries are public, and the p-value attached to $r = .46$ agrees with a correlation of that size in this sample.

1 The question and the headline result

This section answers what Sorenson tested and what happened to that result here. The dissertation asks whether overgeneral autobiographical memory in major depression is tied to abnormal neural activity when a paired image is retrieved. The headline behavioral result is a trend: depressed participants produced fewer specific memories than nondepressed participants. The headline neural result is lower activity in the right middle frontal gyrus during successful retrieval, in the depressed group, together with a positive correlation between that activity and autobiographical specificity. The specificity trend matches the printed moments. The cluster itself remains unchecked, because the images are not public.

Sorenson [1] tested 20 depressed and 19 nondepressed adults, then excluded two nondepressed participants, and analyzed a final sample of 20 and 17. Everyone took the Autobiographical Memory Test and, in the scanner, a paired-associate task that linked concrete nouns to famous faces and places. Table 1 is the comparison. A match means the printed statistic is the value computed from the printed moments, or it lies inside the half-digit bin defined in Section 5. These checks are tail probabilities of printed moments. Agreement is the half-digit bin of those moments. A two-standard-error simulation band, and Bonferroni or Holm on such a band, applies to Monte Carlo rates, which this note does not estimate.

The specificity means are 10.80 (SD 1.64) and 11.82 (SD 1.51). At $n = 20$ and $n = 17$, `simulation/recompute.R` gives $t = -1.9546$, two-tailed $p = .059$. The page prints the unsigned $t(35) = 1.96$, and the prose says the depressed mean is the lower one. The half-digit band on those four moments runs from 1.93 to 1.98 in absolute value, which contains 1.96. The two-tailed p-value of the printed $t(35) = 1.96$ is .0580. The group $F(1, 35) = 3.84$ is the square of that printed t ($1.96^2 = 3.8416$). Figure 1 is the paragraph. The right prefrontal cluster in Table 8 of the dissertation, 220 voxels, was not recomputed. The correlation of that cluster with specificity is $r = .46$, $p < .01$. At $N = 37$, `simulation/recompute.R` gives $p = .0042$ for a correlation of that size. At $N = 36$, allowing the participant whose clarity ratings were missing, $p = .0048$.

negative, and neutral]). For specific memories, there was a trend level significant effect of group, *F*(1,35) = 3.84, *p* = .06, a significant effect of valence, *F*(2,70) = 3.32, *p* = .04, but no interaction of group and valence *F*(2,70) = 0.53, *p* > 0.1. Follow-up post-hoc t-tests indicated that participants generated significantly more specific memories for neutral than for negative cue words (neutral: *M* = 3.97, *SD* = 0.83; negative: *M* = 3.49, *SD* = 0.8; *t*(72) = 2.56, *p* < .05). Participants generated approximately the same number of specific memories for positive and negative cue words (positive: *M* = 3.81, *SD* = 0.97; negative: *M* = 3.49, *SD* = 0.8; *t*(72) = 1.57, *p* > 0.1). Collapsing across valence, there was a trend for depressed participants to generate fewer specific memories than did their nondepressed counterparts (depressed: *M* = 10.80, *SD* = 1.64; nondepressed: *M* = 11.82, *SD* = 1.51; *t*(35) = 1.96, *p* = .058), **see Figure 3**.

Figure 1: Autobiographical Memory Test statistics on dissertation page 28 (PDF page 36), rendered at 600 dpi. The group test is $F(1, 35) = 3.84$, $p = .06$, and the collapsed test is $t(35) = 1.96$, $p = .058$.

2 What we did ourselves, and what we took from others

This section answers which numbers are ours and which are Sorenson's. Sorenson's contribution is the design, the sample, and the printed tables. Our contribution is the recomputation in Table 1, the integer-score checks, the page crops, and the derived effect sizes, which the dissertation does not print. Every later number is one of those three: our calculation, an assumption in Section 5, or a figure taken from the dissertation.

R 4.3.3 [2] is the implementation of the t, F, chi-square, and correlation tails. The dissertation names AFNI, FreeSurfer, nilearn, scikit-learn, and Praat for imaging and for the timing of spoken recall. It does not name the program that produced the behavioral t and F tests, so there is no original version to match. Base R is sufficient: the formulas do not depend on a contributed statistics package. jsonlite 1.8.8 [3] writes the JSON files. The test formulas are base R. Poppler 24.02.0 [4] renders the pages and reads the text layer, because a comma and a period have to be compared on

the page, not only in the extracted string. Python 3.12.3 [5] and Pillow 10.2.0 [6] cut those rasters at the word boxes. pdfTeX, TeX Live 2023 [7], typesets this note. Welch's t, `simulation/recompute.R`, is the second variance formula, used where the Student reading and the printed statistic separate.

The dissertation file is the Stanford Digital Repository PDF, 91 pages, sha256 `4debc9f5551d51ae00a5b6987c3c572a9c7221f6ab631928558c8c1f6b8414d6`. It is not in git. The crops in `paper/figures/` are the page images this note relies on.

3 What the recomputed tests say

This section answers which printed tests follow from the printed moments. Table 1 is that comparison. The paragraphs under the table give one reason for each row. Moments, degrees of freedom, and p-values are in `simulation/out/claims.json`, written by `simulation/recompute.R`.

Table 1: Published claims and our calculations. Unsigned t statistics are compared in absolute value. The prose of the dissertation gives the direction.

Claim	Published	Ours	Verdict
Specific memories, group t	$t(35) = 1.96$, $p = .058$	$t = 1.955$, $p = .059$; band 1.93 to 1.98	match
Specific memories, group F	$F(1, 35) = 3.84$, $p = .06$	$1.96^2 = 3.842$; $p = .058$	match
Valence error degrees of freedom	$F(2, 70)$	$(3 - 1)(37 - 2) = 70$	match
Valence main effect	$F(2, 70) = 3.32$, $p = .04$	$p = .042$; F not recomputed	partial
Group by valence	$F(2, 70) = 0.53$, $p > 0.1$	$p = .591$; F not recomputed	partial
Neutral versus negative cues	$t(72) = 2.56$, $p < .05$	$t = 2.533$, $p = .013$; band 2.46 to 2.60	match
Positive versus negative cues	$t(72) = 1.57$, $p > 0.1$	$t = 1.548$, $p = .126$; band 1.49 to 1.61	match
Gender	$\chi^2(1, N = 39) = 0.20$, $p > 0.1$	Pearson 0.208, $p = .65$	close
Age	$t(37) = 0.08$, $p > 0.1$	$t = -0.080$, $p = .94$	match
Years of education	$t(37) = 1.12$, $p > .1$	$t = -1.120$, $p = .27$	match
Handedness	$\chi^2(1, N = 39) = 0.98$, $p > 0.1$	2×3 Pearson 0.975, df 2, $p = .61$	partial
Shipley vocabulary	$t(36) = 0.55$, $p > .1$	0.554 at 19/19; 0.557 at 20/18	close
Beck depression inventory	$t(35) = 10.94$, $p < 0.001$	10.94 at 20/19 (df 37); 10.36 at 20/17 (df 35)	partial
Positive mood, stated sizes	$t(35) = 1.81$, $p = .08$	$t = -1.804$, $p = .080$; band to -1.809	close
Negative mood, stated sizes	$t(35) = 4.5$, $p < 0.001$	$t = 4.414$, $p = 9.3 \times 10^{-5}$; band 4.40 to 4.43	no match
Both mood scales, $n = 19$ and 18	$t(35) = 1.81$ and $t(35) = 4.5$	$t = -1.811$, $p = .079$; $t = 4.516$, $p = 6.9 \times 10^{-5}$	match
In-scanner clarity, group t	$t(35) = 0.17$, $p > 0.1$	$t = 0$ at the printed means; band -0.49 to 0.49	match

Claim	Published	Ours	Verdict
Clarity, later hit versus miss	$t(36) = 13.08$, $p < 0.001$	$p = 3.0 \times 10^{-15}$; feasible $\lvert t \rvert$ in $[6.7, 29.7]$	partial
Subsequent-memory F denominator	$F(1, 105) = 379.55$, $p < 0.001$	$37 \times 3 - 6 = 105$; p of the printed F is < 0.001	partial
Reward F, decimal comma	$F(1, 35) = 2{,}97$, $p = 0.09$	$p = .094$ for $F = 2.97$; p near 0 for $F = 297$	match
Reward, paired follow-up	$t(36) = 1.80$, $p = 0.08$	$p = .080$; feasible $\lvert t \rvert$ in $[0.38, 4.06]$	partial
Specific hits, group F	$F(1, 35) = 0.01$, $p > 0.1$	$p = .92$; F not recomputed	partial
Reaction-time degrees of freedom	$F(1, 36)$ three times	$p = .28$, 1, .65; the hits tests use df 35	partial
General hits	$t(35) = 1.57$, $p > 0.1$	$t = 1.569$, $p = .126$	match
ROI voxel count	$t(35) = 0.07$, $p > 0.1$	$t = 0.075$, which rounds to 0.07; $p = .94$	match
Encoding classification	$t(35) = 1.26$, $p > 0.1$	$t = 1.237$, $p = .22$; band 1.14 to 1.33	match
Reinstatement above chance	$t(36) = 13.54$, $p < .01$	$t = 13.65$, p near 0; band 13.52 to 13.77	match
Reinstatement, group t	$t(35) = 1.89$, $p = 0.073$	$t = 1.842$, $p = .074$; band 1.79 to 1.89	close
General hits with specificity	$r = -.36$, $p < 0.05$	$p = .029$ at $N = 37$	match
Classifier with specificity	$r = -0.33$, $p < 0.05$	$p = .046$ at $N = 37$; $p = .049$ at $N = 36$	match
Right middle frontal gyrus with specificity	$r = .46$, $p < .01$	$p = .004$ at $N = 37$; $p = .005$ at $N = 36$	match
Two near-zero correlations	$r = .03$ and $r = -0.03$, $p > 0.1$	$p = .86$ and $p = .86$ at $N = 37$	match

The specificity t row is the headline. The moments at 20 and 17 produce 1.95, and 1.96 is inside the half-digit band. The specificity F row is that same contrast: the printed F is the square of the printed t, and its p-value prints as .06 where the t prints as .058. The valence degrees-of-freedom row is the within-subject error of a three-level factor in a two-group sample of 37, which is 70. The valence F row and the interaction row agree in the tail probability of the printed F, and the F itself is not determined by the three marginal means. The two cue-contrast rows use an independent-samples t on 37 and 37, which is what $t(72)$ states. Reading the negative-cue standard deviation as 0.80 rather than as the one-decimal 0.8, both printed t values sit in the half-digit band.

The gender row is Pearson's chi-square on the printed counts 13, 7, 11, and 8, without the continuity correction. The statistic is 0.208. Yates's correction, the alternative tried in Section 4, gives 0.016. The age and education rows use the recruited sizes 20 and 19, because $t(37)$ is n_1+n_2-2 for those sizes, and the absolute values round to the printed statistics. The depressed means are the lower ones, so the signed statistics are negative. The handedness row is the 2×3 table of right, left, and ambidextrous, whose Pearson statistic 0.975 rounds to the printed 0.98, on 2 degrees of freedom. The printed degrees of freedom are 1. Collapsing ambidextrous into "not right" gives 0.17, which is not 0.98.

The Shipley row has two nearby splits that share $t(36)$. Equal sizes 19 and 19 give $t = 0.554$, which rounds to 0.55. The integer-consistent sizes 20 and 18 give $t = 0.557$, which rounds to 0.56. Section 4 records which of those means can be integer totals. The Beck row splits the same way across samples: 10.938 at the recruited 20 and 19, and 10.360 at the final 20 and 17. Welch's formula gives 11.20 and 11.16 on those two splits. The positive-mood row at the stated 20 and

17 is 1.804, and the half-digit band reaches 1.809, just short of the printed 1.81. The p-value .080 prints as .08. The negative-mood row at those sizes is 4.414. The half-digit band is 4.401 to 4.427, and the printed 4.5 lies outside it. The inequality $p < 0.001$ holds for 4.414 ($p = 9.3 \times 10^{-5}$). The mood row at 19 and 18 is the sample that is integer-consistent for both scales and that reproduces both printed t statistics on 35 degrees of freedom. That sample size is an assumption we supplied. It is not a size the dissertation states.

The clarity row treats the two printed means 3.9 and 3.9 as exact and gets $t = 0$. The printed 0.17 lies inside the one-decimal band on those means. The later-hit row checks the printed $t(36) = 13.08$: its p-value is far below .001, and the marginal moments allow a paired t between 6.7 and 29.7. The implied correlation of hit and miss clarity, if the printed t and the printed marginal standard deviations are exact, is 0.55. The subsequent-memory denominator row is the arithmetic $37 \times 3 - 6 = 105$, derived in Section 6. The F of 379.55 was not recomputed. The reward F row reads the printed comma as a decimal comma, because $F(1, 35) = 2.97$ has $p = .094$, which prints as 0.09, and $F = 297$ has a p-value near 0. Figure 2 is that mark. The reward follow-up is feasible: the implied correlation of high-reward and low-reward hit counts is 0.93, inside $(-1, 1)$, and the p-value of $t(36) = 1.80$ is .080. The specific-hits F row agrees in the tail and does not recompute F, because the group means are not printed. The reaction-time row agrees in the three tail inequalities and records that those tests are printed on 36 degrees of freedom while the hits tests in the same paragraph are on 35.

The general-hits row and the voxel-count row round to the printed t at 20 and 17. The encoding-classification row and the above-chance reinstatement row lie in the one-decimal percentage bands. The group reinstatement row is close: the percentages give $t = 1.842$ and $p = .074$, the printed p is 0.073, and the printed $t = 1.89$ has two-tailed $p = .067$. The value 1.89 is the top of the percentage band, so it is reachable, and it is not the t that has $p = 0.073$. Figure 3 is the sentence. The four correlation rows recompute the tail from the printed r and from $N = 37$, or from $N = 36$ where the missing clarity ratings might remove one person. They do not recompute r.

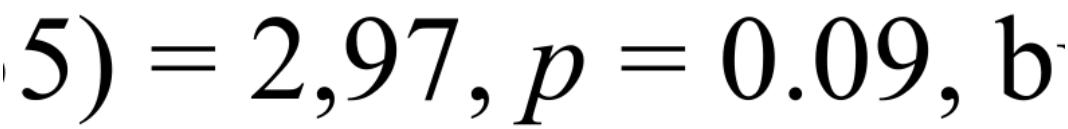


Figure 2: The reward statistic on dissertation page 29 (PDF page 37). The mark between 2 and 97 is a comma. The p-value on the same line uses a period.

Our derived effect sizes, which the dissertation does not print, use the pooled standard deviation from `simulation/recompute.R`. For specific memories, depressed minus nondepressed, Cohen's d is -0.645 and Hedges's g is -0.631. The 95% confidence interval for the mean difference is -2.08 to 0.04, which includes 0. For the Beck inventory at the recruited sizes, $d = 3.50$. At the final sizes, $d = 3.42$. For negative mood at 20 and 17, $d = 1.46$. At 19 and 18, $d = 1.50$. For general hits, $d = 0.518$. For the group difference in reinstatement accuracy, $d = 0.607$. The Hedges factor is $1 - 3/(4\,\mathrm{df} - 1)$.

on a 5-point scale). Across all participants, classification accuracy for retrieval trials was lower than classification accuracy for encoding trials, but was still well above chance (M = 67.5 % SD = 7.8%; chance = 50%; $t(36) = 13.54$ p <.01). Counter to hypotheses, depressed participants showed evidence of *greater* cortical reinstatement at a trend level (depressed: M = 69.5%, SD = 7.0%; nondepressed: M = 64.9%, SD = 8.2%; $t(35) = 1.89$, $p = 0.073$). Intriguingly, across all

Figure 3: Cortical-reinstatement accuracies on dissertation page 30 (PDF page 38). The above-chance test is $t(36) = 13.54$. The group test is $t(35) = 1.89$, $p = 0.073$.

4 What gave us trouble, and how it was resolved

This section answers which gaps closed after the reading, the sample size, the variance formula, and the page image were checked, in that order.

4.1 The Beck statistic and the degrees of freedom

The printed claim is one test, $t(35) = 10.94$. The moments and the recruited sizes 20 and 19 produce $t = 10.938$ on 37 degrees of freedom. The moments and the final sizes 20 and 17 produce $t = 10.360$ on 35 degrees of freedom. Welch's t is 11.199 at 20 and 19, and 11.163 at 20 and 17. The second variance formula does not land on 10.94. Figure 4 is the row. The gap that remains is that no one sample size among these reproduces both the statistic and the degrees of freedom. Both readings give $p < 0.001$.

BDI	M = 26.34, SD = 9.51	M = 1.84, SD = 2.24	t(35) = 10.94, p < 0.001
PANAS: positive	M = 22.89, SD = 8.39	M = 27.78, SD = 8.01	t(35) = 1.81, p = .08
PANAS: negative	M = 16.89, SD = 4.91	M = 11.28, SD = 1.96	*t(35) = 4.5 p < 0.001*
# Specific	M = 10.80, SD = 1.64	M = 11.82, SD = 1.51	t(35) = 1.96, p = .058

Figure 4: Beck and PANAS rows of dissertation Table 1 (PDF page 54). The Beck t is printed on 35 degrees of freedom. The negative-mood t is printed as 4.5.

Integer Beck totals make the same split visible. A 21-item inventory scored 0 to 3 has integer totals from 0 to 63. `simulation/recompute.R` finds no integer total whose average rounds to 26.34 at $n = 20$, and none at $n = 19$. The nondepressed mean 1.84 is the average of the integer 35 at $n = 19$, and it is not an integer average at $n = 17$. A prorated Beck score would fall outside that integer check. The dissertation does not say that any Beck form was prorated.

4.2 The mood means

At the stated final sizes, the positive-mood t is 1.804 against the printed 1.81, and the negative-mood t is 4.414 against the printed 4.5. The negative-mood half-digit band excludes 4.5. PANAS totals are integers from 10 to 50 when each of ten items is scored 1 to 5. Neither printed mood mean is an integer average at 20, and the nondepressed means are not integer averages at 17. Both depressed means are integer averages at 19 (totals 435 and 321). Both nondepressed means are integer averages at 18 (totals 500 and 203). Those sizes sum to 37, so the Student degrees of freedom are 35, which is the printed degrees of freedom. On that split the positive-mood t is -1.811 and the negative-mood t is 4.516, which prints as 4.5. Turning the split off, and using 20 and 17, returns the gap in the negative-mood row. The $p < 0.001$ conclusion holds on both splits.

4.3 The comma in the reward statistic

The text layer and the 600 dpi page both show $F(1, 35) = 2{,}97$, with a comma, beside $p = 0.09$, with a period. `paper/figures/crop_pages.py` cuts that mark, and Figure 2 is the cut. $F(1, 35) = 2.97$ has $p = .0936$. $F(1, 35) = 297$ has $p = 1.1 \times 10^{-18}$. The p-value selects the reading 2.97. The F itself is not recomputed from trial outcomes. The follow-up paired $t(36) = 1.80$ has $t^2 = 3.24$, so it is a different contrast from this F.

4.4 Handedness and the gender hundredth

Figure 5 shows the counts. The gender calculation in `simulation/recompute.R` is 0.20783. Ordinary two-decimal rounding of that value is 0.21. The page prints 0.20. The continuity-corrected statistic is 0.016, so the page is the uncorrected statistic reported a hundredth lower. The inequality $p > 0.1$ holds ($p = .65$). The extra parenthesis after 0.20 is on the page. The handedness counts are 17, 2, and 1 against 17 and 2. The three-category Pearson statistic is 0.975, on 2 degrees of freedom, and the page prints 1 degree of freedom. Expected counts in the ambidextrous column are about one half, so the chi-square tail is a rough approximation there. The inequality $p > 0.1$ holds for the three-category test ($p = .61$) and for the printed statistic on 1 degree of freedom ($p = .32$).

Gender	13 Female, 7 Male	11 Female, 8 Male	$X^2(1,39) = 0.20), p > 0$
Age	$M = 30.35$, $SD = 9.22$	$M = 30.58$, $SD = 8.72$	$t(37) = 0.08, p > 0.1$
Years of Education	$M = 15.95$, $SD = 2.09$	$M = 16.79$, $SD = 2.58$	$t(37) = 1.12, p > .1$
Handedness	17 Right, 2 Left, 1 Ambidextrous	17 Right, 2 Left	$X^2(1,39) = 0.98, p > 0.$

Figure 5: Gender, age, education, and handedness in dissertation Table 1. The gender line prints $\chi^2(1, 39) = 0.20$ with an extra parenthesis. The handedness line prints $\chi^2(1, 39) = 0.98$.

5 Assumptions

This section answers which choices the dissertation does not state and which we supplied. Each subsection is one choice. A sample size or a degrees of freedom that the page states is not in this list.

5.1 Two-sided tails

The page prints inequalities and a few exact p-values, and it does not say whether the tail is one-sided. Every p-value in Table 1 is two-sided. `simulation/recompute.R` uses $2\,P(T \geq |t|)$. One-sided tails would halve those p-values and would still meet the printed inequalities for the rows whose conclusion is $p < .05$, $p < .01$, or $p < 0.001$.

5.2 Half a unit in the last printed digit

A printed moment is treated as any value that rounds to it. A mean or standard deviation printed to two decimals may move by 0.005. A one-decimal number may move by 0.05. `simulation/recompute.R` evaluates the Student t on that box. The negative-cue standard deviation is printed as 0.8. The cue-contrast rows also report the tighter box that reads it as 0.80, and the printed t values sit in that tighter box as well. Percent accuracies printed to one decimal use the wider box.

5.3 Integer instrument totals

The granularity check assumes the usual integer totals: Autobiographical Memory Test counts from 0 to 15, and from 0 to 5 within a valence; Beck totals from 0 to 63; PANAS totals from 10 to 50; Shipley vocabulary from 0 to 40; general hits from 0 to 96. Age uses the stated inclusion window, 18 to 50 years. Education uses the stated floor of 12 years and an upper bound of 30 years that we supplied. The minimum standard deviation in `simulation/recompute.R` is the most equal integers with the required sum. For every row whose mean is an integer average, the printed standard deviation is above that minimum. The output is `simulation/out/grim.json`.

5.4 Mood sample of 19 and 18

The mood row that matches both printed t statistics uses 19 depressed and 18 nondepressed participants. The dissertation states the final analysis sample as 20 and 17. The 19-and-18 split is the nearby split on which both PANAS means are integer averages and both Student statistics print as the page prints them. It is recorded as a separate row so that the stated-size calculation stays visible.

5.5 Correlation sample size

The correlation lines do not print a degrees of freedom. The tails use $N = 37$, the final sample, and, where the missing clarity ratings could remove the one nondepressed participant, also $N = 36$. Both sizes meet the printed inequalities for $r = -.36$, $r = -0.33$, and $r = .46$.

6 Ambiguities that were resolved

This section answers which readings are fixed by an artifact. Each subsection names the script, the output, the command, and the commit PENDINGCOMMIT.

6.1 Student degrees of freedom

The printed $t(35)$, $t(36)$, and $t(37)$ are the equal-variance degrees of freedom $n_1 + n_2 - 2$, or $n - 1$ for a one-sample or paired t. `simulation/recompute.R` is that formula. Welch's t is the alternative, and for the Beck inventory it does not reproduce 10.94. Output: `simulation/out/claims.json`. Command, from the repository root: `Rscript formalization/overgeneral-memory/simulation/recompute.R`.

6.2 The denominator 105

Figure 6 prints $F(1, 105)$ for the subsequent-memory effect in a three-way layout: group, reward, and later hit or miss. A mixed model with the proper subject-by-memory error would use 35 degrees of freedom at $N = 37$. Thirty-seven participants, each with three within-subject degrees of freedom, give 111. The six within-subject terms are reward, memory, their interaction, and each of those with group. The residual is $111 - 6 = 105$. `simulation/recompute.R` is that difference. The same count is 102 at $N = 36$, so the printed denominator agrees with keeping all 37 final-sample participants in the clarity analysis. Output: the `memory_F_df` entry in `simulation/out/claims.json`. Command: the same `Rscript` invocation.

> miss]), ANOVA was conducted on the average clarity ratings for each participant. This analysis yielded a significant main effect of subsequent memory $F(1,105) = 379.55$ $p < 0.001$, but no other main effects or interactions (All Fs < 1, p > 0.1). A follow up paired t-test indicated that during in-scanner retrieval, participants reported remembering word-image pairs that were subsequently remembered during the post-scan retrieval test with higher clarity than they did those image pairs that were subsequently forgotten (subsequent hits: $M = 4.6$; $SD = 0.46$; subsequent misses: $M = 3.28$, $SD = 0.73$; t(36) = 13.08, p < 0.001).

Figure 6: Clarity analyses on dissertation page 28 (PDF page 36). The subsequent-memory effect is printed as $F(1, 105) = 379.55$. The paired follow-up is $t(36) = 13.08$.

6.3 The reward comma

The page image and the text layer agree on a comma in 2,97. The p-value 0.09 agrees with $F = 2.97$ and disagrees with $F = 297$. The crop is `paper/figures/comma-inset.png`, from `paper/figures/crop_pages.py`. Command, from the repository root: `python3 formalization/overgeneral-memory/paper/figures/crop_pages.py -pdf` followed by the local path of the dissertation PDF.

6.4 Pearson versus Yates

The gender counts determine 0.208 without continuity correction and 0.016 with it. The page sits a hundredth from the uncorrected statistic. The continuity-corrected statistic is 0.016. Output: the `gender` entry in `simulation/out/claims.json`.

7 Small differences that are not replication failures

This section answers which printed details differ by a hundredth, a rounding step, or a mark on the page, and whether that difference is a failure of the test. None of the items in this list is a failure of the conclusion the page draws from that test.

The center of the specificity t is 1.95, one step from the printed 1.96, and 1.96 is inside the half-digit band. The square of the center is 3.82, and the printed F is 3.84, the square of 1.96. The gender statistic 0.208 is printed as 0.20, with an extra parenthesis after the number. The Shipley t at the integer-consistent sizes is 0.557, printed as 0.55. The clarity means are both printed as 3.9, and the exact-3.9 t is 0, while 0.17 is inside the one-decimal band. The encoding-classification center is 1.24 against the printed 1.26, inside the percentage band. The above-chance reinstatement center is 13.65 against the printed 13.54, inside the percentage band. The group reinstatement center is 1.84 with $p = .074$, against $t = 1.89$ and $p = 0.073$. The voxel-count t is 0.075, which rounds to the printed 0.07. The methods heading "Cortical Reinstatemet" and the word "furisform" in the classification paragraph are spelling on the page. Table 2 prints a 21-voxel cluster at the boundary of the stated rule "larger than 21 voxels." That boundary is a wording detail of a cluster list we did not recompute.

8 What remains unresolved

This section answers which printed claims stay open, and what was tried on them.

The right prefrontal cluster, and every other cluster table, stays open. The dissertation's univariate results are AFNI `3dMEMA` maps. No image, no bucket file, and no table of participant-level region betas is in the repository record or in the places in Section 9. The correlation $r = .46$ was checked only as a tail probability. The same limit covers encoding maps, reward maps, and the classifier accuracies as neural quantities: the printed percentages were checked as moments, and the images were not reclassified.

The Beck row stays split. Recruited sizes reproduce the t and final sizes reproduce the degrees of freedom. Integer totals reproduce the nondepressed mean at the recruited size and do not reproduce the depressed mean 26.34. The item scores would close this. Welch's t was the second formula, and it did not close it.

The reaction-time degrees of freedom stay one above the hits tests in the same paragraph. $F(1, 36)$ is the equal-variance error for a two-group sample of 38, or $N - 1$ for a sample of 37. The dissertation does not say that a removed participant was put back for the reaction times. The three F values were not recomputed, because the reaction times are not printed. Their tail inequalities hold.

The valence F, the interaction F, the specific-hits F, and the reward F stay as tail checks. Cell means and the within-subject correlations are not printed. The paired clarity t and the paired reward t are feasible given the marginal moments, and they are not identified by those moments.

The exclusion sentence and the clarity degrees of freedom describe different samples. The text says one nondepressed participant lacked clarity ratings and was left out of analyses that used those ratings. The printed clarity tests use $t(35)$, $t(36)$, and $F(1, 105)$, which agree with $N = 37$. The denominator arithmetic in Section 6 uses that N. Which of the two the clarity means actually used is open. A superscript 1 follows the PANAS sentence in the participants section, and the file contains no footnote text for that mark, so whatever it qualified is not in the PDF we read.

GitHub code search did not finish. The last attempt returned HTTP 429. That search is recorded as unfinished in Section 9, not as a completed absence.

9 Code and source availability

This section answers where the dissertation, the code, and the images were sought on 29 October 2026 (UTC). Table 2 is that ledger. "Not available" is not the summary: the PDF was obtained, and several later rows did not obtain a second file.

Table 2: Places checked for the dissertation, for code, and for images.

Place checked	UTC date	Result
Dissertation PDF, Stanford stacks https://stacks.stanford.edu/file/cv880bv1650/Sorenson_Dissertation_To_Submit-augmented.pdf	2026-10-29	Obtained. 91 pages, 3,233,141 bytes, sha256 `4debc9f5551d51ae00a5b6987c3c572a9c7221f6ab631928558c8c1f6b8414d6`. The text names no code deposit.
PURL and Cocina record https://purl.stanford.edu/cv880bv1650	2026-10-29	One file, the PDF. Download is world. License CC BY-NC 3.0.
SearchWorks https://searchworks.stanford.edu/view/11391835	2026-10-29	Bot challenge page. No catalog body. The stacks PDF was already in hand.
ProQuest dissertation search for the title	2026-10-29	HTTP 302, fifty redirects, no file.
OpenAlex author `A5007153948` and a title search	2026-10-29	The author record lists one 2019 arXiv paper and does not list this dissertation. No DOI for the dissertation on that record.
Crossref, "Memory for novel positive information"	2026-10-29	DOI `10.1080/02699931.2013.866936`, 2014, a different title. Full text not retrieved. Not used for these tests.
OSF node titles for this study	2026-10-29	"overgeneral autobiographical" returned four other papers. "Gotlib memory" and "PARC paired associates" returned no nodes. `/v2/search/` returned HTTP 404.
Zenodo, quoted name "James Eric Sorenson"	2026-10-29	Zero records.
GitHub users, "James Eric Sorenson"	2026-10-29	Zero users. Accounts under the surname Sorenson are other people (a media company, an Eric Sorenson, a Lockheed engineer). Those repositories were not cloned.
GitHub code search	2026-10-29	Not completed. HTTP 429.
GitLab projects, "sorenson autobiographical"	2026-10-29	Empty list.
OpenNeuro `advancedSearch`, depression and autobiographical	2026-10-29	No datasets. A keywords-only autobiographical query returned other studies (`ds003511`, `ds000210`, `ds001618`, and others).
NDA search API and query page	2026-10-29	API path HTTP 404. Query page HTTP 302. No dataset record retrieved.

The dissertation names Praat 5.4.08 for spoken reaction times, AFNI for the univariate models, FreeSurfer for the anatomical segmentation, and nilearn with scikit-learn for the classifier. Those are programs, not a deposit of this study's scripts or images. We did not run them.

10 Next steps

This section answers what would close the open rows. The participant-level Autobiographical Memory Test scores, Beck and PANAS item totals, and the paired-associate hit counts would decide whether the mood rows are 20 and 17 or 19 and 18, whether the Beck t and its degrees of freedom come from one sample, and whether the paired t values are the printed ones. The AFNI `3dMEMA` buckets, or the right middle frontal gyrus betas that were correlated with specificity, would decide the cluster contrast and the value of $r = .46$, not only its tail. A session log would name the program that printed the behavioral F tests. A finished GitHub code search, after the rate limit, is the remaining public-code check.

11 Artifacts

This appendix answers where each claim lives and which command writes it. The dissertation PDF is local and gitignored. Its sha256 is `4debc9f5551d51ae00a5b6987c3c572a9c7221f6ab631928558c8c1f6b8414d6`.

- All comparison rows. File `simulation/out/claims.json` and `simulation/out/claims.csv`. Command: `Rscript simulation/recompute.R`.
- Integer-score checks. File `simulation/out/grim.json`. The same command.
- Derived d, g, and confidence intervals. File `simulation/out/derived.json`. The same command.
- R version. File `simulation/out/sessionInfo.txt`. The same command.
- Page crops. Directory `paper/figures/`. Command: `python3 paper/figures/crop_pages.py` with `-pdf` set to the local dissertation file.

References

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